Lecture 6.1 – Frequency Response for Motor Systems

Week 6: A Brief Frequency-Domain View

Learning Objectives

By the end of this lecture, you should be able to:

- explain what frequency response means;
- distinguish low-frequency and high-frequency inputs;
- define magnitude ratio and phase shift conceptually;
- interpret a basic Bode magnitude and phase plot;
- relate a first-order time constant to the corner frequency;
- create a frequency-response plot in Python.

1. Why Look at Frequency Response?

So far, most of the course has used step responses.
A step test answers questions such as:

- How quickly does the motor reach a new speed?
- What is the steady-state error?
- Does the response overshoot?

Frequency response asks a different question:

> How does the system respond when the input changes repeatedly at different frequencies?

This is useful for understanding response speed, filtering, and sensitivity to fast measurement changes.

2. Sinusoidal Inputs

A sinusoidal input can be written as

$$u(t) = A\sin(\omega t)$$

where

- A is input amplitude;
- ω is angular frequency in rad/s.

Frequency in hertz is related by

$$\boxed{\omega = 2\pi f.}$$

A low frequency changes slowly.
A high frequency changes quickly.

3. Output Amplitude and Phase

For a stable linear system, a sinusoidal input produces a sinusoidal steady-state output at the same frequency.
The output can have:

- a different amplitude;
- a phase shift.

For example,

$$u(t) = A\sin(\omega t)$$

may produce

$$y(t) = B\sin(\omega t + \phi).$$

The magnitude ratio is

$$\boxed{|G(j\omega)| = \frac{B}{A}.}$$

The phase angle is

$$\boxed{\phi = \angle G(j\omega).}$$

4. What Does Magnitude Mean?

If

$$|G(j\omega)| = 1,$$

the output amplitude equals the input amplitude.

If

$$|G(j\omega)| < 1,$$

the output is attenuated.

If

$$|G(j\omega)| > 1,$$

the output is amplified.

Magnitude is often shown in decibels:

$$\boxed{M_{\text{dB}} = 20 \log_{10} |G(j\omega)|.}$$

For this class, the most important idea is the trend of the magnitude curve rather than hand-calculating many dB values.

5. What Does Phase Mean?

Phase describes timing between input and output.

If phase is

$$0^\circ,$$

input and output are aligned.

If phase is negative, the output lags behind the input.

A motor cannot instantly follow very fast changes, so higher-frequency inputs typically produce more phase lag.

> Magnitude tells us "how much." Phase tells us "how late."

6. Frequency Response of the First-Order Motor Model

Our Week 2 motor model was

$$G(s) = \frac{K}{\tau s + 1}.$$

For frequency response, replace

$$s$$

with

$$j\omega :$$

$$G(j\omega) = \frac{K}{1 + j\omega\tau}.$$

The magnitude is

$$|G(j\omega)| = \frac{K}{\sqrt{1 + (\omega\tau)^2}}.$$

The phase is

$$\angle G(j\omega) = -\tan^{-1}(\omega\tau).$$

You do not need to memorize these equations. Use them to understand the shape of the Bode plot.

7. Low-Frequency Behavior

When

$$\omega\tau \ll 1,$$

the motor has enough time to follow the input.
Then approximately,

$$|G(j\omega)| \approx K$$

and

$$\angle G(j\omega) \approx 0^\circ.$$

At low frequency, the first-order motor responds almost like its steady-state gain K.

8. High-Frequency Behavior

When frequency becomes large,

$$\omega\tau \gg 1.$$

Then the magnitude becomes small and phase approaches

$$-90^\circ.$$

The motor cannot follow rapid changes very well.

This is one reason high-frequency encoder noise should not be allowed to strongly change the PWM command.

9. Corner Frequency

The first-order corner frequency is

$$\boxed{\omega_c = \frac{1}{\tau}.}$$

In hertz,

$$\boxed{f_c = \frac{1}{2\pi\tau}.}$$

At the corner frequency:

$$|G(j\omega_c)| = \frac{K}{\sqrt{2}} \approx 0.707K$$

and the phase is approximately

$$-45^\circ.$$

If $\tau = 0.50$ s,

$$\omega_c = 2 \text{ rad/s}$$

and

$$f_c \approx 0.318 \text{ Hz}.$$

10. Reading a Bode Plot

A Bode plot uses two graphs:

1. magnitude versus frequency;
2. phase versus frequency.

The frequency axis is logarithmic.
For a first-order motor model:

- low-frequency magnitude is nearly constant;
- magnitude decreases after the corner frequency;
- phase moves from approximately $0°$ toward $-90°$.

11. Python Frequency-Response Plot

A first-order frequency response can be plotted directly using NumPy.

```
import numpy as np
import matplotlib.pyplot as plt

K = 0.775
tau = 0.55

w = np.logspace(-2, 2, 500)

G = K / (1 + 1j*w*tau)

mag_db = 20*np.log10(np.abs(G))
phase_deg = np.angle(G, deg=True)
```

Then create separate magnitude and phase plots.

12. Connection to Filtering and Sampling

Frequency response helps explain two ideas from Week 5.

Filtering

A filter reduces fast measurement changes more than slow changes.

Sampling

A digital controller only receives measurements at discrete times, so it cannot accurately respond to changes that occur much faster than the sampling rate.

> The frequency-domain view supports the same practical lesson as the time-domain view: real motors and digital controllers have limited speed of response.

13. What We Are Not Doing

This course uses frequency response only as a brief design and interpretation tool.

We are not requiring:

- hand-drawn Bode plots;
- gain-margin calculations;
- phase-margin calculations;
- Nyquist plots;
- frequency-domain controller design.

14. Check Your Understanding

Answer briefly.

1. What is the difference between magnitude and phase?
2. If $T = 1/f$, is a 10 Hz signal changing faster or slower than a 1 Hz signal?
3. For a first-order motor, does magnitude generally increase or decrease at high frequency?
4. If $\tau = 0.25$ s, what is ω_c?
5. Why is frequency response useful when discussing measurement noise?

Summary

- Frequency response describes response to repeated sinusoidal inputs.
- Magnitude describes amplitude change.
- Phase describes timing shift.
- A first-order motor follows slow inputs better than fast inputs.
- The corner frequency is $\omega_c = 1/\tau$.
- Python can visualize magnitude and phase without hand plotting.